Dhinakaran Thangaraj

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Professional Summary

Entry-level Machine Learning Engineer skilled in deep learning, NLP, and computer vision. Proficient in Python, ML frameworks, and cloud platforms, with experience in model development, deployment, and driving AI-driven solutions.

Education

- **B.Tech in Artificial Intelligence & Data Science** 2021 – 2025
  Jai Shriram Engineering College, Avinashipalayam, India *CGPA: 8.2*
- **Higher Secondary Education** 2020 – 2021
  Government Higher Secondary School, Muthur, India *Score: 82.1%*

Experience

- **Machine Learning Intern** Aug 2024 – Oct 2024
  Krishtec *Coimbatore*
  - **Real-time PCB Inspection**: Engineered a deep learning-based PCB inspection tool using object detection on NVIDIA Jetson Nano, achieving **90% defect detection accuracy** and improving production efficiency
  - **Performance Optimization**: Improved day-to-day production speed and quality assurance through performance optimization
- **Manufacturing Intern** Aug 2022
  Lenovo *Pondicherry*
  - **Operational Support**: Learned company operations and assisted with assembly processes on the production line
  - **Process Documentation**: Supported creation of workflow documentation and basic quality checks to ensure manufacturing standards

Projects

- **Fine-tuning TinyLLaMA for Text Generation Using PEFT and LoRA**
  Python, PyTorch, Hugging Face Transformers, PEFT, LoRA, Kaggle, Weights & Biases
  - **Model Development**: Spearheaded parameter-efficient fine-tuning of TinyLLaMA transformer on CUAD dataset, optimizing performance for text generation tasks
  - **Results**: Achieved final validation loss of **1.121932** and training loss of **1.025000** after 2800 steps, demonstrating effective transfer learning and model optimization
- **Automated Web Development using LLMs & Streamlit**
  Python, Streamlit, Google Gemini Pro, HTML/CSS
  - **Application**: Built interactive app leveraging Google Gemini Pro and Streamlit to auto-generate and edit HTML/CSS from natural language prompts, enabling rapid prototyping
- **InfoQuery AI Using LangChain & DuckDuckGo API**
  Python, LangChain, Streamlit, DuckDuckGo API, Google Gemini
  - **System Design**: Developed LLM-integrated search engine that parses user queries, retrieves data, and summarizes content ethically through Streamlit UI

Technical Skills

- **Programming Languages**: Python, SQL
- **ML/DL Frameworks**: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, PEFT, LoRA
- **MLOps and Cloud**: AWS (SageMaker, Lambda, EC2), Docker, Git, Weights & Biases, Kaggle
- **Others**: Streamlit, Pandas, NumPy
- **Languages**: Tamil (Native), English (Professional)

Publications

- **AI-Based Mock Interview Platform**: IRJMETS (Apr 2025). Authors: **D. Thangaraj**, T. Selvan, R. Ananth, A. Sivaraj. DOI: 10.56726/IRJMETS72440